



AI detects depression

A ML algorithm can detect depression from the speech of a child.
<https://digit.in/june19-44>



Purify salt water

Researchers have come up with a game changing new way to remove salt from water <https://digit.in/june19-45>



MACHINE LEARNING:

A powerful catalyst in drug discovery

Machine learning in drug discovery and development could be the key to getting more accurate results, all while saving time and money.

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r Alexander Fleming discovered the world's first antibiotic, penicillin. We all know that. However,

Fleming discovered the life-saving molecule in 1928, but it only was made available for the masses 15 years later. In this timeframe, millions lost their lives to bacterial infections, even though the solution was already discovered. Identification, testing, approving and mass producing of medical chemicals takes quite some time. Additionally, the medical organisations involved in drug

discovery incur hefty costs during this process. Machine learning (ML) or deep learning could cut the time and cost considerably and prove to be a more accurate method for drug discovery which can help find cures for potentially terminal illnesses.

DRUG DISCOVERY TODAY

The process of drug discovery has become exceedingly deliberate, strictly monitored and analytically demanding. Estimates show that the introduction of a new drug into the market can cost up to \$4 billion and can take a cumbersome 10-15 years.

ML can prove to be highly useful in the field of drug discovery since it can be utilised to cut down the time

spent on it by a large margin. This is done by using the ability of ML to examine large volumes of data (in this case, medical data) to identify several patterns and connections with a higher efficiency rate than humans could produce. A major part of drug discovery is the process of trial and error. ML algorithms could be utilised to mimic and accurately observe the effects of a large number of chemicals on the target through something known as virtual high-throughput screening (the process is done on a computer using machine learning and



Penicillin took 15 years to hit the market after its discovery

deep learning). ML can analyse larger amounts of data and can also record results of studies in an early verifiable manner. This saves time, cuts down efforts and costs and the room for error in the process.

OTHER BENEFITS OF USING ML IN DRUG DISCOVERY

Machine and deep learning also help in this field in lead optimisation. This is where successfully identified chemicals from the very first screenings are explored further using regression models which are mathematical